

HTML and CSS Basics for Web Scraping

Using BeautifulSoup and Selectors

Bidur Devkota, PhD

Embry-Riddle Aeronautical University

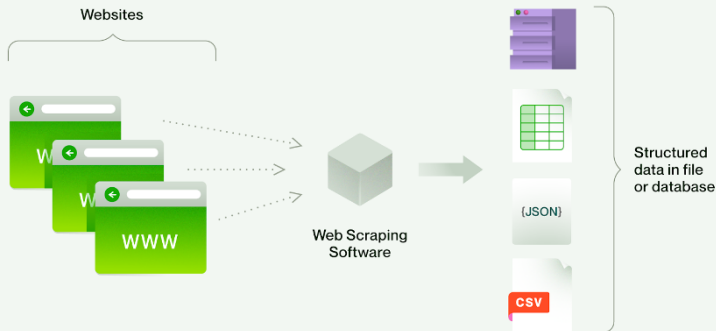
What Is Web Scraping?

- Automated data collection from websites without using APIs
- Programmatically requests and parses HTML content
- Simulates human browsing at scale

Key Reasons for Web Scraping

- Data exists on websites but isn't easily available
- Extract and store information in customized formats
- Process information faster than manual collection

What is web scrapping?



HTTP Request-Response Cycle

Client-Server Interaction

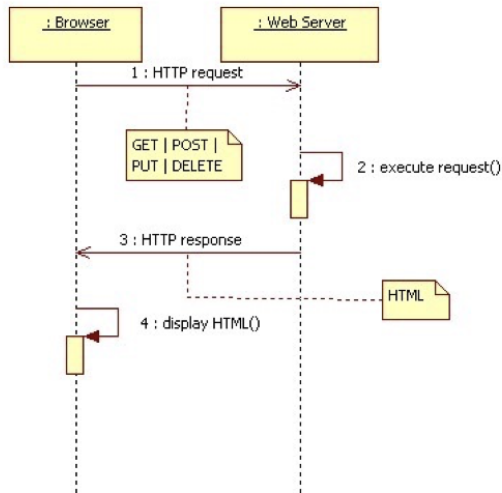
- Client: E.g. Browser
- Server: E.g. Web Server



HTTP Request-Response Cycle

Client-Server Interaction

- Browser sends HTTP request (GET/POST)
- Server responds with HTML content
- Browser renders the response



HTML and CSS Basics for Web Scraping

Basic HTML Structure

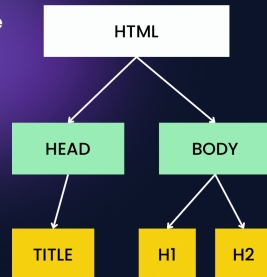
Tree-like Hierarchy

- Root `<html>` element
- `<head>` for metadata
- `<body>` for content
- Tags define structure (`<h1>`, `<div>`, `<p>`, etc.)

Basic HTML Structure

```
<!DOCTYPE html>
<html>
  <head>
    <title> </title>
  </head>

  <body>
    <h1></h1>
    <h2></h2>
  </body>
</html>
```



Basic HTML Structure

- HTML uses `<tag>` to define elements.
- Common tags: `<html>`, `<body>`, `<div>`, `<p>`, `<a>`.

Example:

```
<html>
  <body>
    <p>Hello World!</p>
  </body>
</html>
```


HTML Elements and Scraping Targets

```
example2.html x
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>Web Scraping HTML Example</title>
6 </head>
7 <body>
8
9   <!-- Basic HTML structure -->
10  <p>Hello World!</p>
11
12  <!-- HTML with id and class attributes -->
13  <div id="main-content">
14    <p class="message">Welcome!</p>
15  </div>
16
17  <!-- More HTML tags for CSS selector examples -->
18  <h1>Main Title</h1>
19  <p>This is a normal paragraph.</p>
20  <p class="intro">This is an introduction paragraph.</p>
21
22  <div id="header">
23    <p class="message">Hello from the web!</p>
24  </div>
25
26  <!-- A link and a list -->
27  <a href="https://example.com" class="external-link">Visit Example</a>
28  <ul class="item-list">
29    <li class="item">Apple</li>
30    <li class="item">Banana</li>
31    <li class="item">Cherry</li>
32  </ul>
33
34 </body>
35 </html>
```

Feature	Example in Code
Basic tags	<code><p>Hello World!</p></code>
id usage	<code><div id="main-content"></code>
class usage	<code><p class="message"> Welcome!</p></code>
Multiple CSS targets	<code>.message, #header, .intro, .item</code>
Useful for scraping	elements accessible via modules like BeautifulSoup, Selenium

Commonly Used HTML Tags

Tag	Description
<code><p></code>	Paragraph tag. Defines blocks of text.
<code>
</code>	Line break. Inserts a new line without starting a new paragraph.
<code><div></code>	Division. Used as a container to group elements and apply styles or layout.
<code><a></code>	Anchor tag. Used to create hyperlinks.
<code><table></code>	Creates a table structure. Often used to display tabular data.
<code></code>	Unordered list. Displays a list with bullet points.
<code></code>	List item. Used inside <code></code> or <code></code> tags.
<code><h1></code> to <code><h6></code>	Headings from largest (<code><h1></code>) to smallest (<code><h6></code>).
<code></code>	Embeds an image into the HTML page.
<code></code>	Inline container for text and styles. Similar to <code><div></code> but inline.

HTML Attributes: id and class

- id — uniquely identifies one element.
- class — groups similar elements.

Example:

```
<div id="main-content">  
  <p class="message">Welcome!</p>  
</div>
```

CSS (Cascading Style Sheets) is a language used to describe the presentation of HTML elements.

Why CSS is important:

- Separates content (HTML) from style (CSS).
- Controls layout, colors, fonts, spacing, and more.
- Enables responsive and user-friendly design.
- Makes it easier to maintain large web projects.

Example:

HTML: `<p class="highlight">Important note</p>`

CSS: `.highlight { color: red; font-weight: bold; }`

Ways to Apply CSS in HTML

- **Inline CSS:** Applied directly within an HTML element using the style attribute. `<p style="color:red;">This is red text.</p>`
- **Internal CSS:** Defined inside a `<style>` tag within the `<head>` section of an HTML document. `<style> p color: blue; </style>`
- **External CSS:** Linked from an external stylesheet using the `<link>` tag. `<link rel="stylesheet" href="styles.css">`

CSS Selectors for Scraping

- `tag` — `p`, `div`, `a`, etc.
- `.class` — targets elements by class (e.g., `.message`).
- `#id` — targets elements by ID (e.g., `#main-content`).

Used by tools like BeautifulSoup to locate elements.

Example HTML with CSS

CSS Code

```
body {
  font-family: Arial;
  line-height: 1.6;
  margin: 20px;
}
#main-content {
  background: #f5f5f5;
  border: 1px solid #ddd;
  padding: 15px;
  border-radius: 5px;
}
.message {
  color: #2c3e50;
  font-weight: bold;
}
p.intro {
  font-style: italic;
  color: #34498b;
}
a[href^="https"] {
  color: #e74c3c;
  text-decoration: none;
}
ul.item-list li.item {
  padding: 5px;
  margin-left: 20px;
}
```

HTML Code

```
1      <!DOCTYPE html>
2      <html>
3      <head>
4          <meta charset="UTF-8">
5          <title>Web Scraping HTML Example</title>
6      </head>
7      <body>
8          <p>Hello World!</p>
9          <div id="main-content">
10              <p class="message">Welcome!</p>
11          </div>
12          <h1>Main Title</h1>
13          <p>This is a normal paragraph.</p>
14          <p class="intro">This is an introduction paragraph.</p>
15          <div id="header">
16              <p class="message">Hello from the web!</p>
17          </div>
18          <a href="https://example.com"
19              class="external-link">Visit</a>
20          <ul class="item-list">
21              <li class="item">Apple</li>
22              <li class="item">Banana</li>
23              <li class="item">Cherry</li>
24          </ul>
25      </body>
26      </html>
```

HTML Example

```
example2.html x
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>Web Scrapping HTML Example</title>
6 </head>
7 <body>
8   <!-- Basic HTML structure -->
9   <p>Hello World!</p>
10
11   <!-- HTML with id and class attributes -->
12   <div id="main-content">
13     <p class="message">Welcome!</p>
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21   <div id="header">
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26   <a href="https://example.com" class="external-link">Visit Example</a>
27   <ul class="item-list">
28     <li class="item">Apple</li>
29     <li class="item">Banana</li>
30     <li class="item">Cherry</li>
31   </ul>
32
33 </body>
34 </html>
```

Figure: HTML code

where is the style information?

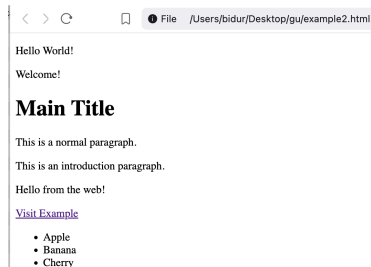


Figure: HTML display in browser

NO STYLE APPLIED!!

CSS applied

```
example2.html X
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>Web Scraping HTML Example</title>
6 </head>
7 <body>
8
9   <!-- Basic HTML structure -->
10  <p>Hello World!</p>
11
12  <!-- HTML with id and class attributes -->
13  <div id="main-content">
14    <p class="message">Welcome!</p>
15  </div>
16
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22  <div id="header">
23    <p class="message">Hello from the web!</p>
24  </div>
25
26  <!-- A link and a list -->
27  <p href="https://example.com" class="external-link">Visit Example</p>
28  <ul class="item-list">
29    <li class="item">Apple</li>
30    <li class="item">Banana</li>
31    <li class="item">Cherry</li>
32  </ul>
33
34 </body>
35 </html>
```

Figure: HTML code

where is the style information?

Hello World!

Welcome!

Main Title

This is a normal paragraph.

This is an introduction paragraph.

Hello from the web!

Visit Example

- Apple
- Banana
- Cherry

Figure: Page displayed with CSS

CSS applied

```
body {
  font-family: Arial;
  line-height: 1.6;
  margin: 20px;
}
#main-content {
  background: #f5f5f5;
  border: 1px solid #ddd;
  padding: 15px;
  border-radius: 5px;
}
.message {
  color: #2c3e50;
  font-weight: bold;
}
p.intro {
  font-style: italic;
  color: #3498db;
}
a[href^="https"] {
  color: #e74c3c;
  text-decoration: none;
}
ul.item-list li.item {
  padding: 5px;
  margin-left: 20px;
}
```

Hello World! p (body)

Welcome! .message

Main Title h1

This is a normal paragraph. p (body)

This is an introduction paragraph. p.intro

Hello from the web! .message

Visit Example a[href^=...

- Apple ul.item-
- Banana ...
- Cherry

Figure: Page displayed with CSS

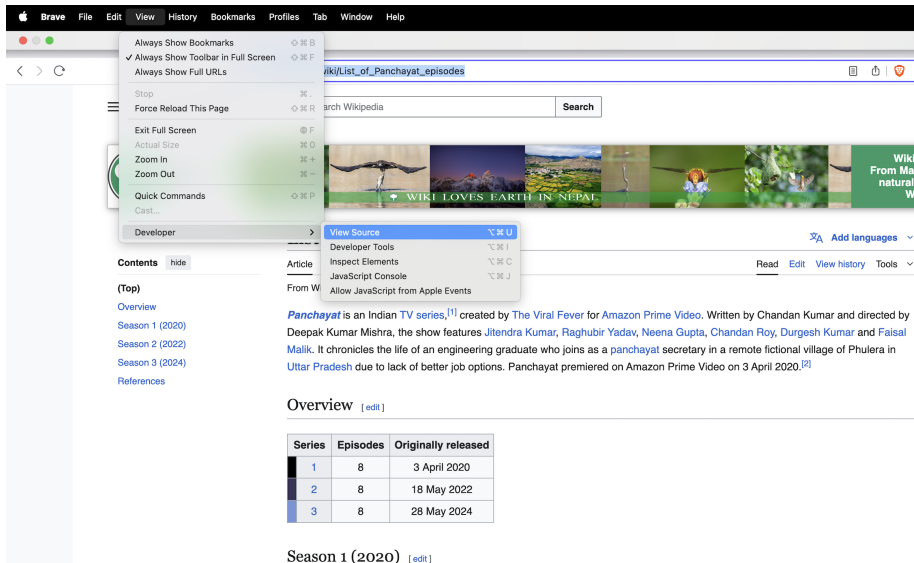
Figure: CSS for the HTML page

Summary of HTML Elements and CSS Selector Used

HTML Element	Selector Used	CSS Properties Applied
Entire page body	<code>body</code>	<code>font-family: Arial, sans-serif; line-height: 1.2;</code>
"Hello World!" paragraph	<code>p</code> (inherits from body)	No specific styles (uses browser defaults)
Main content container	<code>#main-content</code>	<code>background-color: #f5f5f5; border: 1px solid #ccc; border-radius: 5px;</code>
"Welcome!" message	<code>.message</code>	<code>color: #2c3e50; font-weight: bold;</code>
Main title (h1)	<code>h1</code>	Browser default styles
Normal paragraph	<code>p</code>	Browser default styles
Introduction paragraph	<code>p.intro</code>	<code>font-style: italic; color: #3498db;</code>
"Hello from the web!" message	<code>.message</code>	<code>color: #2c3e50; font-weight: bold;</code>
External link	<code>a[href^="https"]</code>	<code>color: #e74c3c; text-decoration: none;</code>
List items	<code>ul.item-list li.item</code>	<code>padding: 5px; margin-left: 20px;</code>

HTML Source view

Any browser can show HTML code of webpage.



The screenshot shows a Brave browser window with the 'View' menu open, highlighting 'View Source'. The background page is a Wikipedia article for 'Panchayat', which is an Indian TV series. The page includes a search bar, a 'Wiki Loves Earth in Nepal' banner, and a table of contents. The 'Overview' section is currently selected, displaying a table with series information.

Series	Episodes	Originally released
1	8	3 April 2020
2	8	18 May 2022
3	8	28 May 2024

HTML inspect element

Hover the mouse cursor on different part of the page and find that different content are organized in different level hierarchically like in a tree. Clicking on anything to focus on a specific part.

W List of Mayday episodes - Wik... x +

< > ↺

en.wikipedia.org/wiki/List_of_Mayday_episodes

📄 🔒 ⚠️

24	10	11 February 2024	14 April 2024
25	11	2 February 2025 ^[2]	26 May 2025

Episodes

Note: Episodes are ordered by their production number, not by their original air date.

table.wikitable.plainrowheader

s.wikiepisodetable

810 × 1314

No. overall	No. in series	Title	Incident	Original release date
1	1	"Unlocking Disaster"	United Airlines Flight 811	10 September 2003

On 24 February 1989, part of the right-side fuselage of United Airlines Flight 811 rips off, ejecting nine people from the aircraft and causing an explosive decompression. The flight later lands safely at Honolulu without any more loss of life. It was later determined that an electrical short circuit caused the cargo door to open.

Note: This accident is covered again in the Season 24 episode "Terror Over The Pacific".

Type of aircraft: Boeing 747-100

2	2	"Racing the Storm"	American Airlines Flight 1420	3 September 2003
---	---	--------------------	-------------------------------	------------------

On 1 June 1999, American Airlines Flight 1420 tries to land at the Little Rock National Airport during a storm, but overruns the runway, killing 11 people. The crash was caused by the crew not arming the spoilers during their pre-landing checks.

Note: Stephen Bogaert, the narrator for the Canadian version, also played First Officer Origel.

Type of aircraft: McDonnell Douglas MD-82

Elements Console Sources Network Performance Memory Application Security Lightb...

<DOCTYPE html>
<html class="client-js vector-feature-language-in-header-enabled vector-feature-language-in-main-pag...
ader-disabled vector-feature-page-tools-pinned-disabled vector-feature-toc-pinned-clientpref-1 vector-fe...
ature-main-menu-pinned-disabled vector-feature-limited-width-clientpref-1 vector-feature-limited-width-c...
ontent-enabled vector-feature-custom-font-size-clientpref-1 vector-feature-night-mode-enabled skin-theme...
-clientpref-0" lang="en" dir="ltr">
 <head>
 </head>
 <body class="skin-responsive skin-vector skin-vector-search-vue mediawiki ltr sitedir-ltr mw-hide-emp...
ty-elt no-8 m-subject page-List_of_Mayday_episodes routage-list_of_Mayday_episodes skin-vector-2022...
action-view-uis-dialog-sticky-hide vector-below-page-title">
 Jump to content
 <div class="vector-header-container"></div> <!-- Flex -->
 <div class="mw-page-container">
 <div class="mw-page-container-inner">
 <div class="vector-sitenotice-container"></div>
 <div class="vector-column-start"></div>
 <div class="mw-content-container">
 <main id="content" class="mw-body">
 <div class="mw-body-header vector-page-titlebar no-font-mode-scale"></div> <!-- Flex -->
 <div class="vector-page-toolbar vector-feature-custom-font-size-clientpref-excluded"></div>
 <div class="vector-column-end no-font-mode-scale"></div>
 <div id="bodyContent" class="vector-body ve-init-mw-desktopArticleTarget-targetContainer" aria-label="body">
 <div class="vector-body-before-content"></div>
 <div id="contentSub"></div>
 <div id="mw-content-text" class="mw-body-content">
 <div class="mw-content-ltr mw-parser-output" lang="en" dir="ltr">
 <p></p>
 <p></p>
 <p></p>
 <style data-mw-deduplicate="TemplateStyles:R806046705"></style>
 <div class="toclimit-2"></div>
 <div class="mw-heading mw-heading3"></div>
 <div style="display:block;overflow:scroll"></div>
 <p></p>
 <div class="mw-heading mw-heading2"></div>
 <p></p>
 <div class="mw-heading mw-heading3"></div>
 <div class="mw-heading mw-heading3"></div>
 <table class="wikitable plainrowheaders wikiepisodetable" style="width:100%; -- s0 -->
 <tbody>
 <tr>
 <td></td>
 </tr>
 </tbody>
 <table data-mw-deduplicate="TemplateStyles:R1237879425"></table>
 <div class="mw-heading mw-heading3"></div>
 <table class="wikitable plainrowheaders wikiepisodetable" style="width:100%; -->
 <tbody>
 <tr>
 <td></td>
 </tr>
 </tbody>
 </table>
 <div class="mw-body-content" id="mw-content-ltr mw-parser-output" table.wikitable.plainrowheaders.wikiepisodetable>
 </div>
 </div>
 </div>
 </div>
 </div>
 </div>
 </div>
 </div>
 </body>
</html>

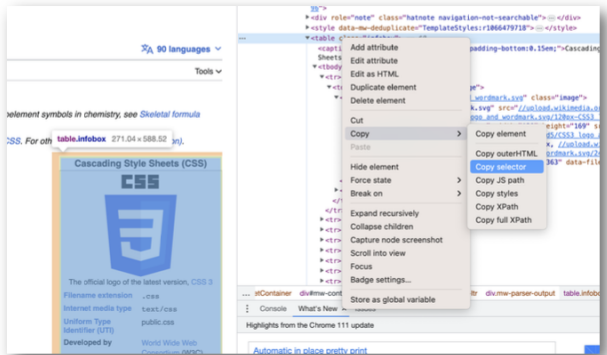
Bidur Devkota, PhD (Embry-Riddle Aeronaut HTML and CSS Basics for Web Scraping

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HTML inspect element

CSS can help to locate different items.

- CSS Selector of the shaded area:
- `#mw-content-text > div.mw-parser-output > table.infobox`
- In JavaScript:
- `document.querySelector("#mw-content-text > div.mw-parser-output > table.infobox")`



BeautifulSoup is a Python library used for parsing and extracting data from HTML documents.

Key Features:

- Parses raw HTML or XML into a navigable tree structure.
- Returns results as Python objects (e.g., strings, lists, dictionaries).
- `find()` gets first match, `find_all()` gets all
- Search by tag name, attributes, or both
- Methods can be chained for nested searches

find() vs find_all()

find()

- Returns first match
- Returns single Tag object
- Returns None if not found

find_all()

- Returns all matches
- Returns ResultSet (list-like)
- Empty list if none found

Syntax

```
soup.find('tag')           # First <tag>
soup.find_all('tag')       # All <tag> elements
```


Common Usage Patterns

By Tag Name

```
soup.find('h1')           # First <h1>
soup.find_all('p')        # All <p> tags
```

By Attributes

```
soup.find(id='header')    # Element with id="header"
soup.find_all(class_='item') # class="item" elements
```

Combined Search

```
soup.find('div', class_='product')
```

Practical Example

```
# HTML: <ul id="products"><li class="item">Laptop</li>...</ul>

products = soup.find('ul', id='products')
items = products.find_all('li', class_='item')

for item in items:
    print(item.text)    # Prints: Laptop...
```

Remember

- class_ (with underscore) not class
- Methods return Tag objects with same methods
- Chain calls: soup.find().find_all()

BeautifulSoup

```
from bs4 import BeautifulSoup
import requests

# Get HTML content
url = "https://example.com"
response = requests.get(url)
soup = BeautifulSoup(response.text, 'html.parser')
```

```
[ ] import requests

# URL to scrape
url = 'http://www.webscrapingfordatascience.com/basichttp/'

# Send GET request
r = requests.get(url)

# Print the raw HTML response
print(r.text)
```

➞ Hello from the web!

Assignment 1: Web Scraping

Objective: Scrape and analyze episode data from Wikipedia.

Task Steps:

- 1 Scrape episode data from:
`https://en.wikipedia.org/wiki/List\_of\_Panchayat\_episodes`
- 2 Use `requests` and `BeautifulSoup` to extract episode text.
- 3 Dump the scraped data into a file.
- 4 Reload the data from the saved file.
- 5 Perform basic text preprocessing, e.g. Stop word filtering.
- 6 Count word frequencies and display the top 10 most frequent words.
- 7 Generate a **word cloud** from the cleaned text.

Play with our class Example @ Colab

Thank You!

References and Resources:

- *Web Scraping for Data Science* – Online Book
- *Lecture Notes: CS540 - Database & Information Retrieval*, Dr. Yongxin Liu, Embry-Riddle Aeronautical University, Daytona Beach, FL
- BeautifulSoup Documentation
- Requests Library Documentation
- WordCloud GitHub Repository
- Wikipedia: List of Panchayat Episodes